CSE-330

Practice Script-23

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Practice Problems : Chapter 3

1. A function is given by $f(x) = x\,e^{-3x} + x^2$. Now answer the following up to five significant figures.
   a. Approximate the derivative of $f(x)$ at $x_0 = 2$ with step size $h = 0.1$ using the central difference method.
   b. Calculate the upper bound of the truncation error of $f(x)$ at $x_0 = 2$ using $h = 0.1$ using the central difference method.
   c. Compute $D^{(1)}_{0.1}$ at $x_0 = 2$ using Richardson extrapolation method and calculate the relative error.
2. During the derivation process, we explored the first-order Richardson extrapolated derivative using $(h \to \frac{h}{2})$,

$$D_h^{(1)} \equiv f'(x_0) - \frac{h^4}{480} + O(h^6)$$

   a. Using $h \to \frac{h}{2}$, derive the expression for $D^{(2)}_{\frac{h}{2}}$, which represents the second-order Richardson extrapolation.
   b. Now, starting from the definition of $D_h^{(1)}$ and applying $h \to \frac{h}{3}$, derive the expression for $D_h^{(1)}$.

3. Consider the function $g(x) = \ln(x)$,
   a. Approximate the derivative of $g(x)$ at $x_0 = 1.5$ with step size $h = 0.2$ using the central difference method up to 6 significant figures.
   b. Approximate the derivative of $g(x)$ at $x_0 = 1.5$ with step size $h = 0.2$ using the forward difference method up to 6 significant figures.
   c. Calculate the truncation error of $g(x)$ at $x_0 = 1.5$ using $h = 0.1, 0.01, 0.001$ in both the central difference and forward difference methods.
   d. Compute Compute $D^{(1)}_{0.2}$ at $x_0 = 1.5$ using Richardson extrapolation method up to 6 significant figures and calculate the truncation error. $D^{(1)}_{0.2}$ at $x_0 = 1.5$ using Richardson extrapolation method up to 6 significant figures and calculate the truncation error.
4. Given the function $f(x) = e^{-2x}$:
   a. Approximate the derivative of $f(x)$ at $x_0 = 0.5$ with step size $h = 0.2$ using the forward difference method up to 6 significant figures.
   b. Approximate the derivative of $f(x)$ at $x_0 = 0.5$ with step size $h = 0.2$ using the central difference method up to 6 significant figures.
   c. Calculate the truncation error of the derivative approximation at $x_0 = 0.5$ using $h = 0.1, 0.01, 0.001$ for both the forward and central difference methods.
   d. Compute $D^{(1)}_{0.2}$ at $x_0 = 0.5$ using Richardson extrapolation up to 6 significant figures and calculate the truncation/relative error.

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5. Consider the following data table::

x	1.1	1.2	1.3
f(x)	-0.57941	-0.90730	-1.2807

a. Using the above data, compute f'(1.2) using the central difference method.
b. For the interval [1.1, 1.3], compute the error bound (truncation error) if the above data is generated by the function, $f(x) = x\cos(x) - x^2\sin(x)$.
c. Also compute the actual error.

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Ans to the Qno: 1

$f(x) = xe^{-3x} + x^2$

$f'(x) = x(-3e^{-3x}) + (e^{-3x}) + (2x)$

$f'(x) = -3xe^{-3x} + 2x + e^{-3x}$

$f''(x) = -3xe^{-3x} + 2 + -3x(-3e^{-3x}) + e^{-3x}(-3)$

$= 9xe^{-3x} + 2 - 6e^{-3x}$

$f'''(x) = 27e^{-3x} + -27xe^{-3x}$

$f^4(x) = 27[x\cdot(-3e^{-3x}) + e^{-3x}]$

$= -81xe^{-3x} + 27e^{-3x}$

$f^5(x) = -81[x(-3e^{-3x}) + e^{-3x}(1)] - 81e^{-3x}$

$= +243xe^{-3x} - 81e^{-3x} - 81e^{-3x}$

$= 243xe^{-3x} - 162e^{-3x}$

a) $f'(x) = \frac{f(x+h) - f(x-h)}{2h}$

$f'(2) = \frac{f(2+0.1) - f(2-0.1)}{2(0.2)}$

$f(2+0.1) = 4.41$

$f(2-0.1) = 3.62$

$f'(2) = \frac{4.41 - 3.62}{0.2}$

$= 3.95$ ✗

$= 3.98$ (Be more accurate) ✓

b) Error $= \frac{h^2}{6} f'''(\epsilon)$ ~~$+\frac{h^2}{6} f'''(\epsilon)$~~

$= \frac{f'''(\epsilon)h^2}{6}$

$= \frac{+(0.1)^2}{6} f'''(2)$ ~~$+\frac{(0.1)^2}{6} f'''(2)$~~

$= \frac{27(2)e^{-3(2)}}{6} \times (0.1)^2$

$= 2.231 \times 10^{-4}$ ~~$+1.115 \times 10^{-4}$~~

c) $D_h = f'(x) + \frac{f'''(x)h^2}{3!} + \frac{f^v(x)h^4}{5!}$ — (i)

$D_{\frac{h}{2}} = f'(x) + \frac{f'''(x)h^2}{3! \times 4} + \frac{f^v(x)h^4}{5! \times 16}$ — (ii)

$16 \times (ii) - (i) \Rightarrow 3f'(x) + \frac{f^5(x)h^4}{5!}\left(\frac{1}{4} - 1\right)$

$4D_{\frac{h}{2}} - D_h = 3f'(x) - \frac{3}{4}\frac{f^5(x)h^4}{5!}$

$\therefore D_h^{(1)} = \frac{4D_{\frac{h}{2}} - D_h}{3} = f'(x) - \frac{1}{4}\frac{f^5(x)h^4}{5!}$

$D'(0.1) = f'(2) - \frac{1}{4}\frac{f^5(2)(0.1)^4}{5!} \Rightarrow 3.987606 - \frac{1}{4}\frac{0.803 \times (1 \times 10^{-4})}{120} = 3.9876$

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Ans to the Qno: 3

$g(x) = \ln x$

$x_0 = 1.5$

$h = 0.2$

a) $f'(x) = \dfrac{f(x_0+h) - f(x_0-h)}{2h}$

$= \dfrac{f(1.7) - f(1.3)}{2(0.2)}$

$= \dfrac{\ln(1.7) - \ln(1.3)}{0.4}$

$= 0.670660$

b) $f'(x) = \dfrac{f(x+h) - f(x)}{h}$

$= \dfrac{f(1.7) - f(1.5)}{0.2}$

$= 0.625816$

c) Truncation error formula for forward difference $= \dfrac{f''(\xi)h}{2}$

Truncation error formula for central difference $= \dfrac{f'''(\xi)h^2}{6}$

For forward difference,

$x_0 = 1.5$

$h = 0.1, 0.01, 0.001$

$g(x) = \ln x$

$g'(x) = \dfrac{1}{x}$

$g''(x) = -\dfrac{1}{x^2}$

$g''(x) = -x^{-2}$

$g'''(x) = -(-2x^{-3})$

$g'''(x) = 2x^{-3}$

$g'''(x) = \dfrac{2}{x^3}$

$g^4(x) = 2(-3x^{-4})$

$= -\dfrac{6}{x^4}$

$= -6x^{-4}$

$g^5(x) = -6(-4x^{-5})$

$= 24x^{-5}$

$= \dfrac{24}{x^5}$

Ans to the Qno:4

$f(x) = e^{-2x}$

a) $f'(x) = \frac{f(x_0+h) - f(x_0)}{h}$

$= \frac{f(0.7) - f(0.5)}{0.2}$

$= \frac{e^{-2(0.7)} - e^{-2(0.5)}}{0.2}$

$= -0.606412$

b) $f'(x) = \frac{f(x_0+h) - f(x_0-h)}{2h}$

$= \frac{f(0.7) - f(0.3)}{0.4}$

$= -0.755537$

$f^0(x) = e^{-2x}$

$f'(x) = -2e^{-2x}$

$f''(x) = 4e^{-2x}$

$f'''(x) = -8e^{-2x}$

$f^4(x) = -8(-2e^{-2x})$

$= 16e^{-2x}$

$f^5(x) = -32e^{-2x}$

c) Truncation error,

at Forward difference $= \frac{f''(\xi)h}{2}$

at Central difference $= \frac{f'''(\xi)h^2}{6}$

$x_0 = 0.5$

When $h = 0.1$,

Truncation error at forward difference $= \frac{4e^{-2(0.5)} \times 0.1}{2}$

$= 0.07358$

Truncation error at central difference $= \frac{-8e^{-2(0.5)} \times 0.1}{6}$

$= -0.049051 = -.491 \times 10^{-1}$

Ans to the Qno: 5

a) $f'(1.2) = \frac{f(x_0+h) - f(x_0-h)}{2(0.1)}$

$= \frac{f(1.3) - f(1.1)}{0.2}$

$= \frac{-1.2807 + 0.97941}{0.2}$

$= -3.51$

When $h = 0.01$

Truncation error at forward difference $= \frac{4e^{-2(0.5)} \times 0.01}{2}$

$= 7.358 \times 10^{-3}$

Truncation error at central difference $= \frac{-8e^{-2(0.5)} \times (0.01)^2}{6}$

$= -4.91 \times 10^{-5}$

When $h = 0.001$,

Truncation error at forward difference $= \frac{4e^{-2(0.5)} \times 0.001}{2}$

$= 7.358 \times 10^{-4}$

Truncation error at central difference $= \frac{-8e^{-2(0.5)} \times (0.001)^2}{6}$

$= -4.91 \times 10^{-7}$

d) $D_h = f'(x) + \frac{f^3(x)h^2}{3!} + \frac{f^5(x)h^4}{5!}$ — (1)

$$D_{\frac{h}{2}} = f'(x) + \frac{f^3(x)h^2}{3!} \times \frac{1}{4} + \frac{f^5(x) \times h^4}{5! \times 16} \quad - (11)$$

$4(11) - (1) =$ ~~$3f'(x) - \frac{3}{4}\frac{f^3(x)h^2}{3!} + -\frac{3}{4}f^5(x)$~~

$= 3f'(x) + 0 + \frac{f^5(x)h^4}{5!}(-1)$

$= 3f'(x) - \frac{3}{4}\frac{f^5(x)h^4}{5!}$

$$\frac{4D_{\frac{h}{2}} - D_h}{3} = f'(x) - \frac{1}{4}\frac{f^5(x)h^4}{5!}$$

$$D_h^{(1)} = f'(x) - \frac{1}{4}\frac{f^5(x)h^4}{5!}$$

$D_{0.5}^{(1)}$ at $x_0 = 0.5 = f'(0.5) - \frac{1}{4} \cdot \frac{f^5(0.5)h^4}{5!}$

$= -2e^{-2(0.5)} - \frac{1}{4}\frac{-32e^{-2(0.5)} \times (0.2)^4}{5!}$

$= -0.235720$

Truncation error $= -\frac{1}{4}\frac{f^5(x)h^4}{5!}$

$= \frac{-1}{4}\frac{-32e^{-2(0.5)} \times (0.2)^4}{5!}$

$= 3.92405 \times 10^{-5}$

Ans to the Qno: 2

a) $D_h^{(1)} = f'(x_0) - \frac{h^4}{480} + O(h^6)$ — (1)

$D_{\frac{h}{2}}^{(1)} = f'(x_0) - \frac{1}{480} \times \frac{h^4}{16} + O(h^6)$ — (11)

$16(11) - (1) \Rightarrow$

$15f'(x_0) - \frac{1}{480}h^4 + \frac{1}{480}h^4 + O(h^6)$

$= 15f'(x_0)$

$\therefore \frac{16D_{\frac{h}{2}}^{(1)} - D_h^{(1)}}{15} = 15f'(x_0) + O(h^6)$

$\therefore D_h^{(2)} = f'(x_0) + O(h^6)$

$D_{\frac{h}{2}}^{(2)} = f'(x_0) + O(h^6)$

b) $D_h = f'(x) + \frac{f^3(x)h^2}{3!} + \frac{f^5(x)h^4}{5!}$ — (1)

$D_{\frac{h}{3}} = f'(x) + \frac{f^3(x)}{3!} \times \left(\frac{h}{3}\right)^2 + \frac{f^5(x)}{5!} \times \left(\frac{h}{3}\right)^4$

$D_{\frac{h}{3}} = f'(x) + \frac{f^3(x)}{3!} \times \frac{h^2}{9} + \frac{f^5(x)}{5!} \times \frac{h^4}{81}$ — (11)

$9(11) - (1) \Rightarrow 8f'(x) + 0 + \frac{f^5(x)h^4}{5! \times 9} - \frac{f^5(x)h^4}{5!}$

$= 8f'(x) - \frac{8}{9}\frac{f^5(x)h^4}{5!} = 9D_{\frac{h}{3}} - D_h$

$D_h^{(1)} = \frac{9D_{\frac{h}{3}} - D_h}{8}$

$D_h^{(1)} = f'(x) - \frac{1}{9}\frac{f^5(x)h^4}{5!}$

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a) Central difference $= \dfrac{f(x_0+h)-f(x_0-h)}{2h}$

$= \dfrac{f(1.3)-f(1.1)}{2(0.1)}$

$= \dfrac{-1.2807-(-0.57941)}{0.2}$

$= -3.50645$

b) Truncation error $= \dfrac{f'''(\xi)h^2}{6}$

$f(x) = x\cos(x) - x^2\sin(x)$

$f'(x) = x(-\sin x)+\cos x(1) - x^2(\cos x) - \sin x(2x)$

$= -x\sin x + \cos x - x^2\cos x - 2x\sin x$

$= -3x\sin x + \cos x - x^2\cos x$

$f'' = -3\left[x(\cos x)+\sin x\right] - \sin x - \left[x^2(-\sin x)+\cos x(2x)\right]$

$= -3x\cos x - 3\sin x - \sin x + x^2\sin x - 2x\cos x$

$f''(x) = -5x\cos x - 4\sin x + x^2\sin x$

$f'''(x) = -5\left[x(-\sin x)+\cos x\right] - 4\cos x + x^2(\cos x)+\sin x(2x)$

$= -5\left[-x\sin x+\cos x\right] - 4\cos x + x^2\cos x + 2x\sin x$

$= +5x\sin x - 5\cos x - 4\cos x + x^2\cos x + 2x\sin x$

$f'''(x) = -9\cos x + 7x\sin x + x^2\cos x$

Max truncation error $= \dfrac{f'''(\xi)h^2}{6}$

$= 0.011355$

(c) $f'(1.2) = -3.514778121$

$f'(1.2) = -3.50645$

Actual error $= -3.514778121 - (-3.50645)$

$= -0.00833$

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